

Set Theory Enderton - Solutions Manual

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Chapter 1

Introduction

Exercise 1

- (a) Both $\in$ and $\subseteq$.
- (b) Only $\subseteq$.
- (c) Only $\subseteq$.
- (d) Only $\in$.
- (e) None works.

Exercise 2

The empty set contains no element, whereas the others both contain something. The second set contains the empty set, but the third does not. Hence, they are pairwise unequal.

Exercise 3

$$a \in \mathcal{P}(B) \Rightarrow a \subseteq B \Rightarrow a \subseteq C \Rightarrow a \in \mathcal{P}(C)$$

where the second implication comes from the transitivity of the subset relation.

Exercise 4

$$x, y \in B \Rightarrow \{x, y\}, \{x\} \in \mathcal{P}(B) \Rightarrow \{\{x, y\}, \{x\}\} \in \mathcal{PP}(B)$$

Exercise 5

The rank of $\{\{\emptyset\}\}$ is 2. The rank of $\{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$ is 3.

Exercise 6

I don't understand the question.

Exercise 7

Fuh naw.

Chapter 2

Axioms and Operations

Exercise 1

Set of integers divisible by 180.

Exercise 2

Let $A = \{\{1\}, \emptyset\}$ and $B = \{\{1\}\}$.

Exercise 3

If $B \in A$, then $\bigcup A$ includes every $x \in B$. Hence, $B \subseteq \bigcup A$.

Exercise 4

$$x \in \bigcup A \Rightarrow x \in X \in A \Rightarrow x \in X \in B \Rightarrow x \in \bigcup B$$

Exercise 5

$$x \in \bigcup A \Rightarrow x \in X \in A$$

Since $X \in A$, then X is a subset of B . Hence,

$$\begin{aligned} x \in X &\subseteq B \\ \Rightarrow x &\in B \end{aligned}$$

Exercise 6

(a)

$$x \in \bigcup \mathcal{P}A \Rightarrow x \in X \in \mathcal{P}A$$

By definition of power set, this means that X is a subset of A . Hence,

$$x \in X \subseteq A$$

$$\Rightarrow x \in A$$

For the other direction, if $x \in A$, then the singleton $\{x\}$ is a subset of A . In other words, $\{x\} \in \mathcal{P}A$. Since $x \in \{x\} \in \mathcal{P}A$, then $x \in \bigcup \mathcal{P}A$.

(b)

$$X \in A \Rightarrow X \subseteq \bigcup A \Rightarrow X \in \mathcal{P} \bigcup A$$

Equality only holds when A is a power set (of some arbitrary set).

Exercise 7

(a)

$$\begin{aligned} X &\in \mathcal{P}A \cap \mathcal{P}B \\ \Leftrightarrow X &\in \mathcal{P}A \text{ and } X \in \mathcal{P}B \\ \Leftrightarrow X &\subseteq A \text{ and } X \subseteq B \\ \Leftrightarrow X &\subseteq A \cap B \\ \Leftrightarrow X &\in \mathcal{P}(A \cap B) \end{aligned}$$

(b)

$$\begin{aligned} X &\in \mathcal{P}A \cup \mathcal{P}B \\ \Rightarrow X &\in \mathcal{P}A \text{ or } X \in \mathcal{P}B \\ \Rightarrow X &\subseteq A \text{ or } X \subseteq B \\ \Rightarrow X &\subseteq A \cup B \\ \Rightarrow X &\in \mathcal{P}(A \cup B) \end{aligned}$$

The crucial step is when: $X \subseteq A$ or $X \subseteq B$ implies $X \subseteq A \cup B$. The reverse is not necessarily true. In fact, to make it true, either $A \subseteq B$ or $B \subseteq A$.

Exercise 8

Suppose for contradiction that X contains every singleton $\{x\}$. Using the Union Axiom, we get $\bigcup X$ which contains every set x , which is impossible.

Exercise 9

I mean, generically, the power set of an element in B will not be a subset of B .

Exercise 10

We know from a previous exercise that $B \subseteq \mathcal{P} \bigcup B$. So, if $a \in B$, then $a \in \mathcal{P} \bigcup B$, hence $a \subseteq \bigcup B$. Now, clearly, every subset of a will be a subset of $\bigcup B$ by transitivity. Formally, $x \subseteq a \Rightarrow x \subseteq \bigcup B$. This is equivalent to $\mathcal{P}a \subseteq \mathcal{P} \bigcup B$. Hence, $\mathcal{P}a \in \mathcal{P} \mathcal{P} \bigcup B$.

Exercise 11

First equation:

$$A = A \cap (B \cup B^c) = (A \cap B) \cup (A \cap B^c) = (A \cap B) \cup (A - B)$$

Second equation:

$$A \cup (B - A) = A \cup (B \cap A^c) = (A \cup B) \cap (A \cup A^c) = A \cup B$$

Note that the complement is not a set. Although it can be avoided, it's just easier to write down and conveys the same idea.

Chapter 3

Relations and Functions